

Project Initialization and Planning Phase

Date	4 July 2026
Team ID	
Project Title	Comprehensive measure of well-being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform national human development analysis using machine learning, boosting speed and accuracy in socio-economic assessments. It tackles the inefficiencies of manual data processing, promising faster regional planning, reduced administrative overhead, and clearer policy decisions. Key features include a machine learning-based well-being index model and real-time, form-driven prediction.

Project Overview	
Objective	The primary objective is to streamline global human development analysis by implementing machine learning techniques, ensuring instant and accurate well-being assessments.
Scope	The project comprehensively evaluates and enhances the socio-economic tracking process, incorporating automated data parsing and an interactive web interface for dynamic model forecasting.

Problem Statement	
Description	Traditional human development tracking is delayed by massive, fragmented multi-decade public datasets and messy formatting, creating severe bottlenecks in manual planning.
Impact	Resolving these issues will significantly speed up regional planning cycles, optimize the distribution of public development aid, and reduce administrative computational friction.

Proposed Solution	
Approach	Deploying machine learning regression algorithms to evaluate socio-economic indicators and instantly calculate a standardized, bounded target well-being score.
Key Features	• Implementation of a predictive machine learning-based index model. • An intuitive, uncluttered web form layout for real-time scenario simulation. • Automated handling of missing variables and spaces in dataset rows.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard Quad-Core CPU (Intel i5/AMD Ryzen 5 or equivalent)
Memory	RAM specifications	8 GB RAM or above
Storage	Disk space for data, models, and logs	256 GB SSD / 500 MB allocated for project data & model binaries
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Data	Source, size, format	Kaggle / United Nations Public Datasets, ~150 KB, .csv format